REVIEW-II

RoadWatch AP — Automatic Road Extraction and New-Road Change Alerting from Multi-Temporal Satellite Images

Smart India Hackathon SIH1564 (NTRO) · KL Capstone CLIMATE-06

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KONERU LAKSHMAIAH EDUCATION FOUNDATION

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INTRODUCTION

- Roads change every season; maps are updated by hand, slowly and incompletely
- Unseen new roads mean planning, logistics and monitoring work from out-of-date evidence
- RoadWatch reads free Sentinel-2 satellite images (10 m, every 5 days), maps every road and alerts on genuinely new ones — with before/after proof
- Built for SIH1564 (NTRO); runs for Andhra Pradesh, Tamil Nadu, Hyderabad, Bengaluru and is being extended to all of India



Pre-monsoon 2022



Pre-monsoon 2024

Same place, two years apart
Vijayawada West Bypass (NH16): the new bridge across the Krishna appears between the two Sentinel-2 seasons — the kind of change RoadWatch alerts on.

PROBLEM STATEMENT & OBJECTIVES

Road networks change constantly and keeping maps current by hand is slow and incomplete, so spatial decisions are made from misaligned, incomplete or weakly validated evidence. Goal: extract roads automatically from satellite images and raise trustworthy alerts for new ones.

O1

Validate the problem

Charter, requirements, risk register, source-to-claim map

Done

O2

Robust road extraction

U-Net on Sentinel-2; frozen spatial holdout; 95 % CIs

Done

O3

Change over time

Graph comparison across 9 seasons, persistence, abstain

Done

O4

Alerts with proof

Scored alerts, evidence chips, flags, live web app

Done

O5

Independent validation

KPIs vs OSM history, negative tests, all-India check

In progress

FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

Functional

FR-1	Ingest Sentinel-2 L2A + OSM labels with provenance and quality gates	Met
FR-2	One-command reproducible training runs (config hash per run)	Met
FR-3	Change detection as a separate, tested subsystem	Met
FR-4	Alert layer: score, evidence, input/output contracts	Met
FR-5	CLI pipeline: s2-build · infer · detect · serve	Met
FR-6	Complete, deployed system — not a notebook	Met

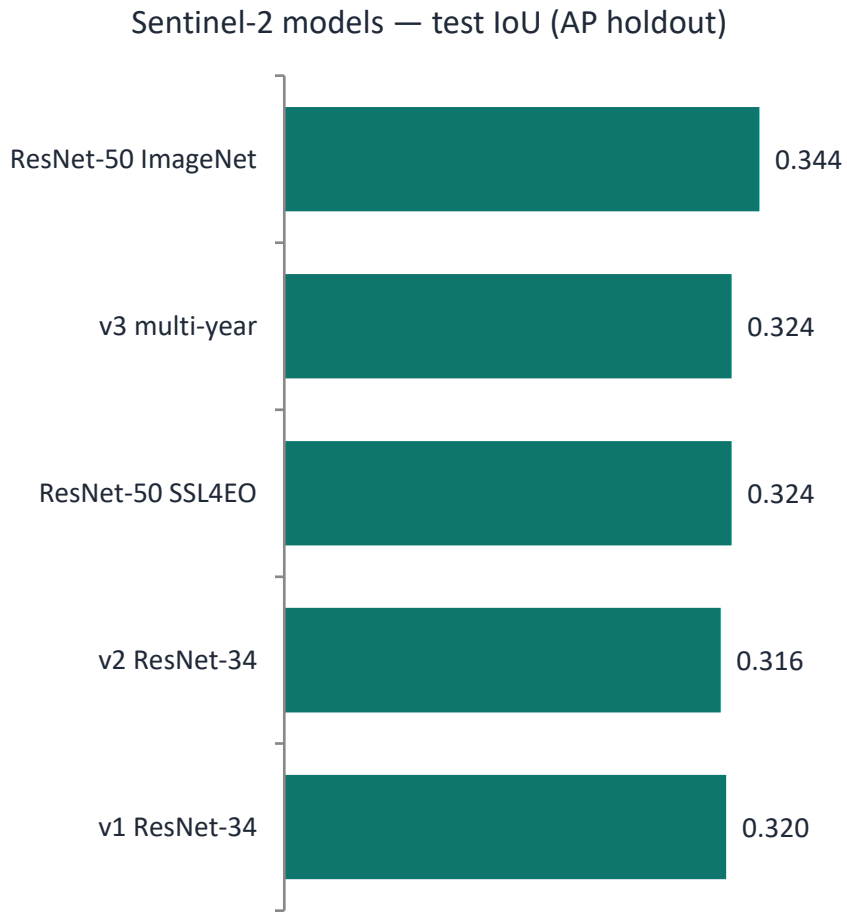
Non-functional

NFR-1	Performance: KPIs with 95 % CIs, resource cost recorded	Partly
NFR-2	Reliability: all negative tests pass (14/14)	Met
NFR-3	CRS, registration, cloud cover explicit and checked	Met
NFR-4	Frozen spatial holdout — no train/test leakage	Met
NFR-5	Git, CI, automated tests, Docker build	Met
NFR-6	Operability: health, degraded mode (abstain), guides	Partly

ALTERNATIVE SOLUTIONS & TRADE-OFF ANALYSIS

Approach	Data	Measured	Decision
Pixel difference (classical)	Same model, 2 dates	6,826 alerts, precision 0.20	Baseline only
U-Net ResNet-34 (Baseline A)	DeepGlobe 0.5 m RGB	Test IoU 0.501	Reference; 0.5 m data not free statewide
Frozen DINOv3-sat ViT-L (Ref. B)	DeepGlobe 0.5 m RGB	Test IoU 0.356	Rejected (below A)
U-Net on Sentinel-2 10 m	Andhra Pradesh, 4 bands	Test IoU 0.320 → 0.344	**Chosen: free, 5-day revisit, statewide**

Graph comparison over 9 seasons cuts reviewer load 16× vs pixel difference; pixel difference with 30 m tolerance stays a strong baseline (see Results).



COMPLETE SYSTEM ARCHITECTURE

Sources

- Sentinel-2 L2A (Planetary Computer)
- OpenStreetMap (Geofabrik, ohsome)
- Esri Wayback (sharp reference)

Kaggle (free CPU / GPU)

- Seasonal composites
- Model training (U-Net)
- Statewide screening runs

Merge & publish

- Merge, stable ids, flags
- Road vector tiles, bundle
- GitHub release + commit

Serve (Render)

- FastAPI: alerts, evidence, tiles
- React + MapLibre web app
- Reviews → training labels

GitHub Actions scheduler
Twice a year (5 Jun, 5 Jan) it builds the new season on Kaggle, re-screens, merges, flags and republishes the site with no manual step. CI runs lint, tests, holdout check and a Docker smoke test on every push.

TECHNOLOGY STACK & DESIGN RATIONALE

Geospatial data

Free, repeatable imagery with recorded provenance

- Sentinel-2 L2A via STAC
- rasterio / GDAL, GeoPandas, Shapely
- Deterministic UTM tile grid

Machine learning

CPU inference anywhere; paired evaluation with CIs

- PyTorch + segmentation-models
- U-Net, ResNet-34 / 50 encoders
- ONNX Runtime for inference

Web & API

Fast map of ~1 M road lines on a 512 MB server

- FastAPI (Python 3.11)
- React, TypeScript, Vite
- MapLibre GL vector tiles, three.js

Ops & quality

Every claim reproducible from a config hash

- Docker on Render
- Kaggle runs, GitHub Actions
- pytest (88 tests), ruff, CI

DETAILED DESIGN & DATA FLOW

1 — Composite
Median of the 8 least-cloudy passes per season, 10 m, 4 bands

→

2 — Segment
U-Net road probability per 5.12 km tile (ONNX)

→

3 — Vectorise
Threshold → skeleton → road graph per season

→

4 — Compare
2022 baseline (25 m buffer) vs every later season

→

5 — Confirm
Seen on ≥ 3 later dates, stability ≥ 0.75; cloudy = abstain

→

6 — Alert
Score, stable id, flags, before/after evidence

Alert score = model confidence × persistence × connectivity × data quality

Stable ids
An alert overlapping an earlier one keeps its id, so reviews stay on the same road

Flags
On a road OSM already had in 2022, or a short line inside dense buildings → listed apart

Quality gates
Mixed CRS or out-of-order seasons are refused; registration error ≤ 1 px

IMPLEMENTATION & UNIT-LEVEL PROGRESS

11,920

alerts published
AP, Tamil Nadu, Hyderabad, Bengaluru

6,449

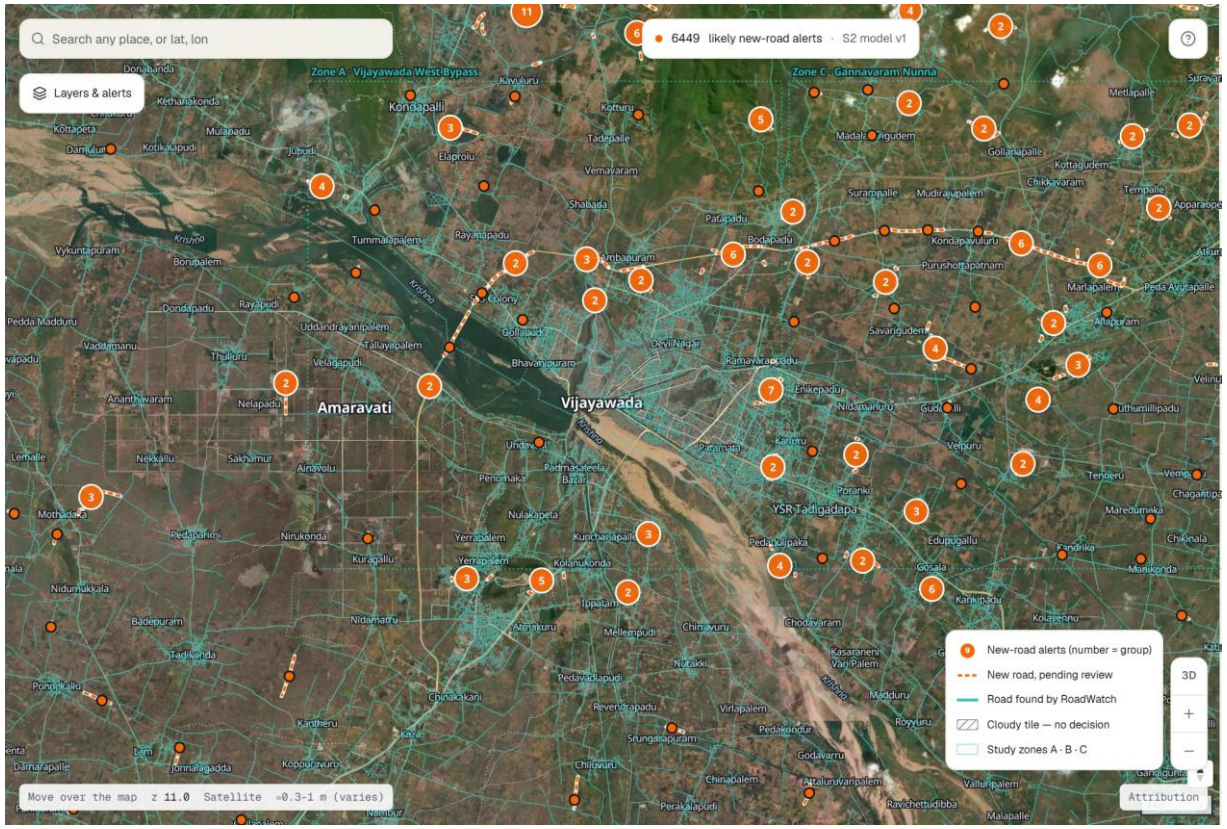
likely new roads
5,471 flagged as probably not new

2.9 lakh km

roads extracted
served as map vector tiles

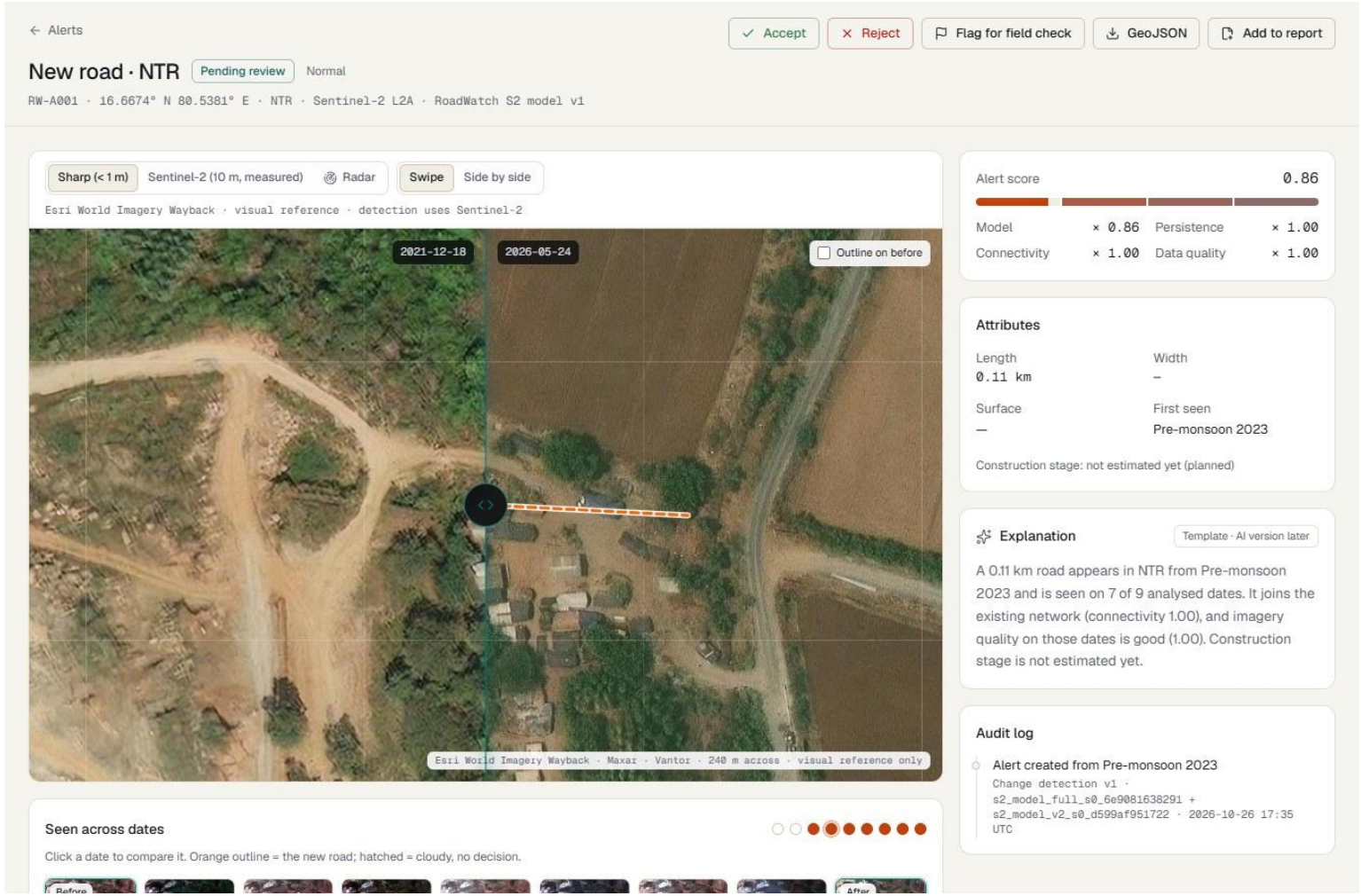
98.7 %

of South India mapped
64,327 of 65,166 tiles; North running



Live map, Vijayawada: teal = roads extracted by RoadWatch, orange = new-road alerts (numbers = clusters)

THE LIVE SYSTEM — EVERY ALERT WITH PROOF



Before / after
Sharp Esri Wayback imagery for reference; detection itself uses Sentinel-2

Seen across dates
Filmstrip of all 9 seasons: seen, absent or cloudy

Accept / reject / flag
Decisions are exported as training labels

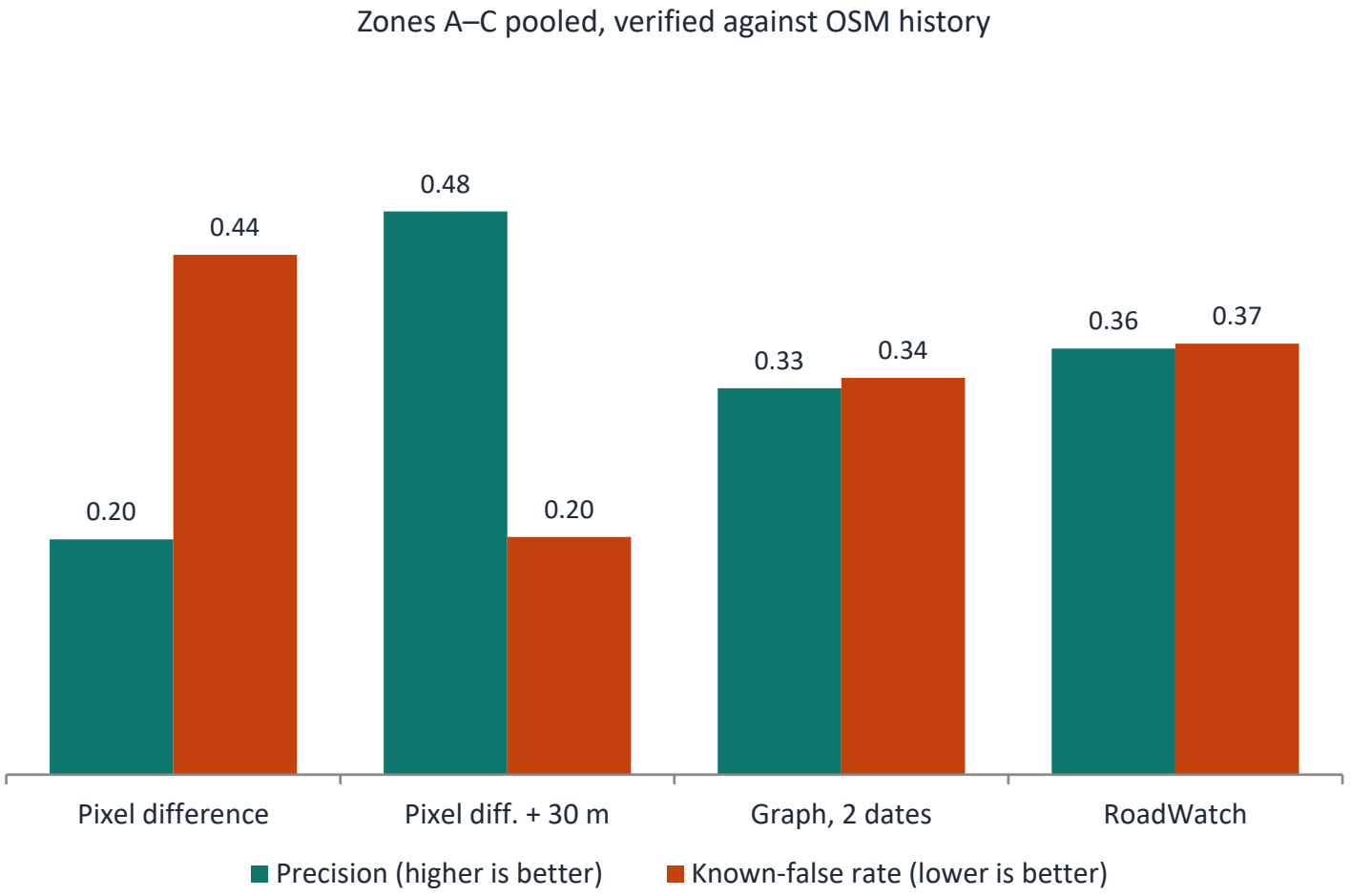
Try it
roadwatch-ap.onrender.com

TESTING & RESULTS

Check	What was measured	Result
KPI-1 Road extraction	Pixel IoU, AP spatial holdout, 95 % bootstrap CI	**0.344 (ResNet-50); v1 0.320 [0.300, 0.338]**
KPI-2 New-road precision	Alerts vs OSM history 2022 → 2026, zones A–C	**0.364 [0.305, 0.427]**
KPI-3 False alerts	Alerts on roads OSM already had in 2022	**0.368 vs 0.444 pixel difference (−17 %)**
KPI-6 Registration	Shift between seasons, 522 tile pairs	**median 0.071 px; 100 % ≤ 1 px**
Negative tests NT-1…NT-5	Occlusion, seasonal tracks, CRS, cloud, shift	**14 / 14 pass**
Engineering	Automated tests + CI (lint, holdout, Docker)	**88 tests pass; site API 3–4 s → 0.34 s**

Honest reading: extraction is measured with confidence intervals and every engineering check passes; change detection beats raw pixel difference, but about 1 in 3 alerts still lies on an old road — those are now flagged and listed apart.

CHANGE DETECTION VS BASELINES



413 **alerts to review**

vs 6,826 from pixel difference — 16× less work

0.364 **precision**

above pixel difference (0.201); below the 30 m-tolerant variant (0.481)

5,471 **alerts flagged**

lie on 2022 OSM roads or are short lines in built-up areas

CONCLUSION & FUTURE SCOPE

Conclusion

- A complete, deployed system: satellite images in, new-road alerts with proof out
- Extraction measured honestly on a frozen holdout (best test IoU 0.344)
- Change detection cuts reviewer load 16× and every alert carries evidence
- Updates itself twice a year — no laptop, no manual step

Future scope

- Finish the all-India map (North half running) and close the remaining gaps
- Multi-year training to cut alerts on old roads (root cause found)
- Sentinel-1 radar for the cloudy monsoon months
- Field validation with district officials; reviews feed retraining

0.344 best test IoU (extraction)

16× fewer alerts than pixel difference

11,920 alerts live, updated twice a year

THANK YOU

Questions?

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